XLVIII National Mathematics Olympiad: 4th round, 18–19 May 1999

Problem 1. The faces of an orthogonal parallelepiped whose dimensions are natural numbers are painted green. The parallelepiped is partitioned into unit cubes by planes parallel to its faces. Find the dimensions of the parallelepiped if the number of cubes having no green face is one third of the total number of cubes.

Problem 2. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of integer numbers such that

$$(n-1)a_{n+1} = (n+1)a_n - 2(n-1)$$

for any $n \geq 1$. If 2000 divides a_{1999}, find the smallest $n \geq 2$ such that 2000 divides a_n.

Problem 3. The vertices of a triangle have integer coordinates and one of its sides is of length $\sqrt{n}$, where n is a square-free natural number. Prove that the ratio of the circumradius and the inradius is an irrational number.

Problem 4. Find the number of all natural numbers $n, 4 \leq n \leq 1023$, such that their binary representations do not contain three consecutive equal digits.

Problem 5. The vertices A, B and C of an acute triangle ABC lie on the sides B_1C_1, C_1A_1 and A_1B_1 of $\triangle A_1B_1C_1$ and $\angle ABC = \angle A_1B_1C_1$, $\angle BCA = \angle B_1C_1A_1$, $\angle CAB = \angle C_1A_1B_1$. Prove that the orthocentres of $\triangle ABC$ and $\triangle A_1B_1C_1$ are equaly remote from the circumcentre of $\triangle ABC$.

Problem 6. Prove that the equation

$$x^3 + y^3 + z^3 + t^3 = 1999$$

has infinitely many integer solutions.

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